

BUSINESS REQUIREMENT DOCUMENT

Project SanskritiSetu: Offline-First Living Cultural Discovery, Last-Mile Transit & Ground-Readiness Engine

DOCUMENT REF	PROBLEM STATEMENT	CLASSIFICATION	VERSION & STATUS
BRD-SIH2026-SS-01	SIH26044 / SIH-TOURIST	Digital Public Infrastructure (DPI)	v1.0 • Approved for Implementation

1. EXECUTIVE SUMMARY & PROBLEM CONTEXT

1.1 Executive Summary

SanskritiSetu is an offline-first, edge-native cultural discovery, last-mile ground navigation, and traveler safety platform designed to decentralize tourism across India. By bridging the gap between national cultural inventory databases and ground-level execution, the platform enables domestic and international travelers to discover date-shifting living heritage, navigate cellular dead zones, verify logistical realities, and engage in middleman-free commerce with rural and Adivasi communities.

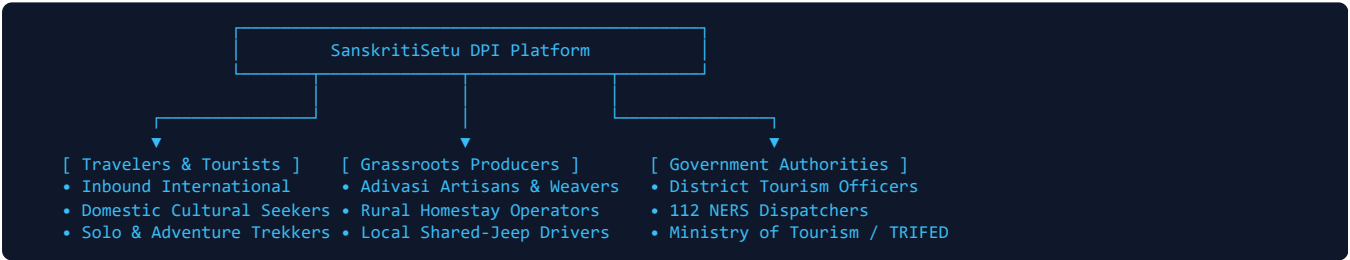
1.2 Problem Context & Industry Baseline

- Severe Footfall Concentration:** According to Ministry of Tourism statistics, over **65% of all domestic tourist visits** are confined to just 5–6 states, while **40%+ of inbound international travelers** never leave the Golden Triangle (Delhi–Agra–Jaipur) and Goa corridor.
- The Living Heritage Discovery Gap:** The Ministry of Culture’s *Mera Gaon Meri Dharohar* (MGMD) initiative has mapped over **6.38 lakh villages** and **2,500+ major traditional fairs**. However, **<1% of these dynamic, date-shifting events** are discoverable on commercial Online Travel Agencies (OTAs) because they follow fluctuating lunisolar (Panchang, Hijri, Saka) calendars rather than fixed Gregorian dates.
- The Ground-Level Execution Void:** Travelers attempting to explore off-circuit rural regions (e.g., Ladakh, the North-East, and Central tribal belts) encounter acute failures:
 - Cellular Blackouts:** High-density cultural pockets often have zero data connectivity, rendering cloud-dependent apps useless.
 - Unmapped Last-Mile Transit:** Absence of scheduled transit, leaving tourists dependent on informal, unscheduled shared jeeps/sumos.
 - Dead-Zone Cash Collapse:** Complete inoperability of UPI/POS payment rails in dead zones without prior cash buffer warnings.
 - Sanitation & Health Blindspots:** Lack of verified intelligence regarding clean drinking water, Western toilets, and medical clinics (antivenom/AMS).
 - Permit Friction:** Unpreparedness for Inner Line Permits (ILP) and Restricted Area Permits (RAP) at physical military/police checkposts.
- Economic Leakage:** **Less than 5% of inbound tourism expenditure** reaches rural or indigenous economies, causing distress out-migration among master artisans, weavers, and folk performers into urban manual labor.

2. BUSINESS OBJECTIVES & SUCCESS METRICS (KPIs)

OBJECTIVE	TARGET KPI	MEASUREMENT BASELINE
Footfall Decentralization	≥40% increase in off-circuit travel routing to tier-3, rural, and tribal circuits within 18 months of pilot deployment.	Ministry of Tourism Regional Footfall Telemetry and State Border Checkpost Logs.
Artisan Economic Upliftment	3x increase in take-home artisan margins by cutting out intermediaries for GI/ODOP and tribal crafts.	Direct-to-consumer transaction ledgers via TRIFED/ODOP nodes.
Rural Homestay Occupancy	≥40% year-round occupancy growth for NIDHI 2.0 / SAATHI verified rural accommodations.	NIDHI 2.0 host check-in dashboards.
Traveler Safety & Zero Dead-Zone Stranding	100% emergency reachability via sub-cellular failover; sub-15 minute response coordination with 112 NERS.	State Emergency Response Operations Centre (EROC) dispatch logs.
Administrative Scalability	Zero portal abandonment by District Tourism Officers (DTOs) via ≤2 minutes/week automated batch verification overhead.	DTO workspace event approval turnaround time logs.

3. STAKEHOLDER ECOSYSTEM & USER PERSONAS



3.1 Primary Personas

- **Persona A: Inbound International / Solo Traveler (Claire, 31):** Seeks authentic cultural experiences (e.g., Nagaland Hornbill, Bastar Madai), but faces severe anxiety regarding physical safety, hygiene, female-friendly sanitation, permit paperwork, and becoming stranded without cash or cellular connectivity.
- **Persona B: Adivasi Artisan / Weaver (Somu, 44, Bastar):** Produces authentic Dhokra metalcraft and Kotpad handloom fabrics; lacks digital literacy and English fluency; exploited by urban middlemen taking 70–80% retail cuts; desires direct footfall to his village workshop.
- **Persona C: Rural Homestay Operator (Stanzin, 38, Ladakh):** Runs a verified village homestay registered on NIDHI; struggles with severe seasonal peaks and 8 months of vacancy; needs a steady stream of culturally respectful visitors aligned with local monastic festivals.
- **Persona D: District Tourism Officer (DTO / State Administrator):** Chronically understaffed; burdened with protocol duties; cannot spend hours vetting crowdsourced portal submissions, but requires accurate crowd management and traveler safety tracking in remote talukas.

4. SCOPE OF THE SOLUTION

4.1 IN-SCOPE (PHASE 1 & PHASE 2) • Pan-India 365-Day Cultural Radar: Real-time temporal and spatial query engine mapping concurrent living festivals, rituals, and weekly haats.	4.2 OUT-OF-SCOPE (DEFERRED ITERATIONS) • Direct hosting of airline Global Distribution Systems (GDS) or running proprietary payment wallets (uses
--	--

- **Astronomical Ephemeris Engine:** Mathematical conversion of lunisolar Panchang, Hijri, and regional solar calendars to Gregorian timelines up to 3 years forward.
- **Ground-Readiness Index (GRI):** Algorithmic scoring of remote locations for cash availability, sanitation, transit, and medical support.
- **Zero-Grid Offline Client Architecture:** Dynamic packaging layer producing self-contained <60 MB regional bundles with offline maps, search, and language inference.
- **Sub-Cellular 2G Binary SMS SOS:** 40-character payload compression routing distress telemetry directly into the national 112 NERS gateway.
- **Two-Tier Administrative Approval Workflow:** Automated anomaly detection pre-screening crowdsourced data for 1-click batch authorization by DTOs.
- **Permit Assistance Module:** Digital application links and auto-generation of checkpoint-ready printable physical registration dossiers for ILP/RAP zones.

standard UPI/banking gateways or cash-buffer advisory).

- Real-time GPS vehicle tracking of informal transit (shared sumos) where cellular signals do not physically exist.
- Physical infrastructure construction (sanitation, roads, or telecommunication towers).
- Sovereign visa issuance or direct border entry clearance bypassing state and federal law enforcement authorities.

5. FUNCTIONAL BUSINESS REQUIREMENTS

5.1 Module 1: Temporal Cultural Discovery & Ephemeris Engine (TCD)

- **BR-TCD-01 (Nationwide Temporal Radar):** The system shall provide an interactive visual timeline and map allowing users to select any calendar date or window (1–30 days) and view all active cultural events across all 28 States and 8 UTs.
- **BR-TCD-02 (Astronomical Conversion):** The system shall mathematically calculate date shifts for non-Gregorian festivals (lunar tithis, nakshatras, Hijri sightings) using high-precision astronomical algorithms (IAU standard) with zero dependency on manual admin updates.
- **BR-TCD-03 (Time-Variant Living POI Schema):** Event schemas shall store temporal boundaries (start date, peak ritual windows, concluding processions, flexibility buffers) rather than static latitude/longitude points alone.

5.2 Module 2: Ground-Readiness Index (GRI) & Logistics Telemetry (GRL)

- **BR-GRL-01 (GRI Multi-Factor Scoring):** The system shall compute and display a transparent Ground-Readiness Index (A to D rating) for every remote destination based on 4 mandatory parameters:
 1. *Cash Liquidity:* Distance to the nearest verified functioning ATM and currency requirements.
 2. *Sanitation & Hygiene:* Presence of clean drinking water and Western-style vs. pit latrines.
 3. *Healthcare Access:* Distance and transit time to the nearest Primary Health Centre (PHC), antivenom storage, or High-Altitude Medical Clinic.
 4. *Road Accessibility:* Paved road vs. unpaved gravel vs. 4x4 requirement.
- **BR-GRL-02 (Dead-Zone Cash Estimator):** The system shall compute an estimated physical cash budget required per traveler per day (incorporating homestay tariffs, local guide fees, food, and craft purchase buffers) and warn the user at the last known urban ATM node.
- **BR-GRL-03 (Transit Lifeline Indexing):** The platform shall index unmapped, informal last-mile transit hubs (e.g., shared sumo stands, weekly market pickup points, tractor trailheads) including typical morning departure windows and localized fare benchmarks.

5.3 Module 3: Zero-Grid Offline Edge Native Stack (ZGE)

- **BR-ZGE-01 (Micro-Bundle Generation):** The backend shall dynamically package geographic districts into downloadable bundles strictly under **60 MB**, containing vector map tiles (.mbtiles), vector embeddings, and localized POI records.
- **BR-ZGE-02 (On-Device Local Search):** The mobile client shall execute offline semantic search and discovery using an embedded vector database (SQLite-VSS) without querying external cloud infrastructure.
- **BR-ZGE-03 (Edge Intelligence & Dialect Translation):** The client shall embed a quantized 1B Small Language Model (SLM) executing via ExecuTorch on device silicon to answer cultural queries and translate phrases across 22 scheduled Indian languages offline.

5.4 Module 4: Traveler Safety & Emergency Response (TSR)

- **BR-TSR-01 (2G Binary SMS SOS Gateway):** In the event of an emergency trigger in a zero-cellular-data zone, the app shall compress current GPS coordinates, timestamp, battery status, and emergency flag into a **40-character binary SMS payload**.
- **BR-TSR-02 (112 NERS Integration):** The telephony bridge shall forward this binary SMS directly into the National Emergency Response Support System (112 NERS) gateway and alert local District Tourist Police checkpoints.
- **BR-TSR-03 (Contextual Cultural Etiquette Cards):** The system shall trigger visual, multi-lingual "Dos and Don'ts" cards based on proximity to sensitive sacred sites, tribal sanctums, or monastic halls (dress codes, photography bans, circumambulation direction) to prevent cultural friction.

5.5 Module 5: Administrative Triaging & DTO Workspace (ATW)

- **BR-ATW-01 (Automated Data Hygiene Filter):** Crowdsourced event submissions from local ambassadors, homestay hosts, and cultural committees shall pass through an automated NLP filter checking for duplicate records, geographic coordinate consistency, and profanity.

- **BR-ATW-02 (One-Click Batch Verification):** The web dashboard for District Tourism Officers (DTOs) shall display pre-vetted, risk-scored event queues allowing 1-click batch approval or rejection, limiting administrative workload to ≤ 2 minutes per week.
- **BR-ATW-03 (Footfall Monitoring & Alerting):** The DTO portal shall visualize real-time download and routing intent heatmaps, triggering carrying-capacity warnings when remote eco-sensitive zones face severe over-tourism risks.

5.6 Module 6: Economic Marketplace & Bureaucracy Integration (EMB)

- **BR-EMB-01 (Middleman-Free Marketplace):** The platform shall enable verified Adivasi artisans, weavers, and ODOP producers to showcase products and on-site masterclasses with 100% direct traveler payout (via cash on-site or direct UPI when connected).
- **BR-EMB-02 (Homestay Accreditation Sync):** The platform shall ingest and cross-reference homestay registration data with the Ministry of Tourism's **NIDHI 2.0 / SAATHI** database to display official safety and verification badges.
- **BR-EMB-03 (Permit Navigator & Physical Form Generation):** For destinations designated under Inner Line Permit (ILP) or Protected/Restricted Area Permit (PAP/RAP) jurisdictions, the system shall provide direct official application links and auto-fill exportable, printable physical PDF registration slips formatted for military/police checkpost logbooks.

6. NON-FUNCTIONAL REQUIREMENTS (NFRS)

6.1 Performance & Hardware Constraints

- **Client App Footprint:** Base application size must not exceed **35 MB** on initial download.
- **Regional Bundle Size:** Individual district offline bundle download sizes must strictly remain **<60 MB**.
- **Edge Inference Latency:** On-device vector similarity search (SQLite-VSS) query response time must be **<250 ms** on mid-tier mobile processors (e.g., Snapdragon 600-series or MediaTek Dimensity equivalent).
- **Battery Consumption:** Continuous background offline GPS location listening in transit mode must consume **<3% battery per hour**.

6.2 Reliability & Offline Availability

- **Zero-Grid Independence:** 100% of core discovery, map navigation, cultural querying, and etiquette guidance features must function without active cellular data once a regional bundle is cached.
- **Graceful Degradation:** The platform must transition automatically between 5G/4G, 2G fallback, and complete offline mode without session dropouts or UI freezes.

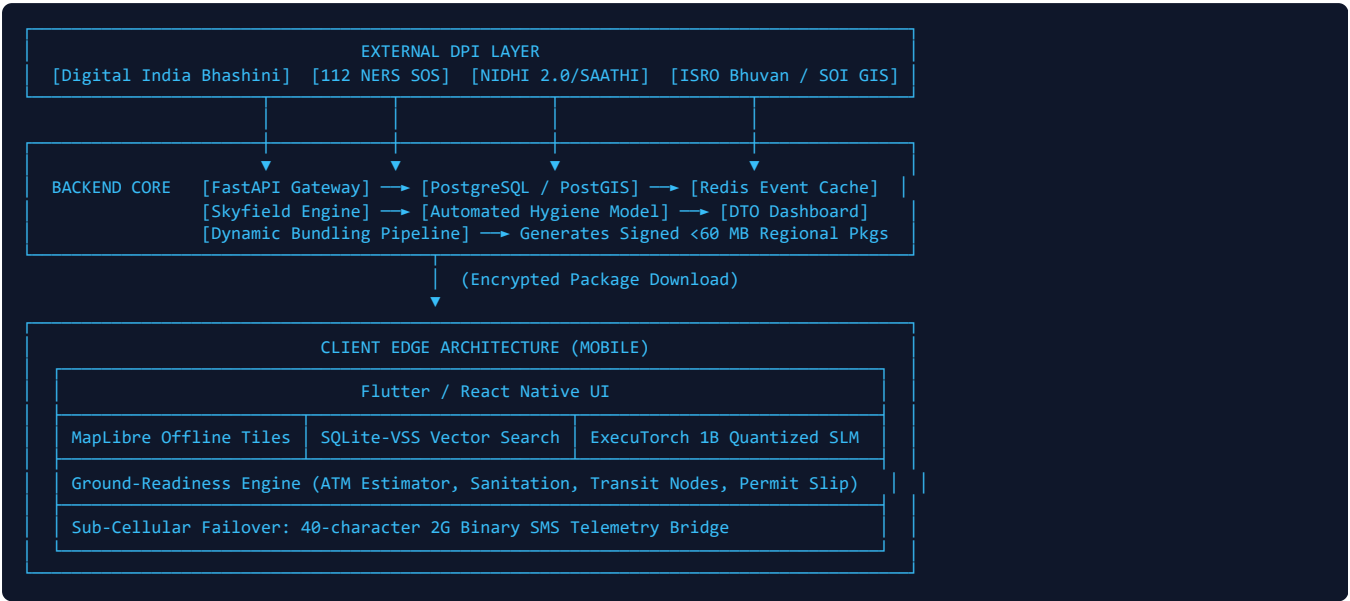
6.3 Data Privacy & Statutory Compliance

- **DPDP Act 2023 Compliance:** Zero persistent background location tracking on remote servers. All telemetry and location profiles must remain localized on the user's phone silicon.
- **Sovereign Data Storage:** All central administrative databases, GIS mapping nodes, and DTO portals must reside strictly within Indian sovereign cloud infrastructure (MeitY-empanelled data centers).

6.4 Security & Access Control

- **Role-Based Access Control (RBAC):** Tiered administrative levels (Super Admin, State Tourism Board, District Tourism Officer, Verified Local Host).
- **Cryptographic Signing:** Regional offline data bundles must be cryptographically signed via **Ed25519 keys** to prevent malicious tampering, route poisoning, or fake listing injections.

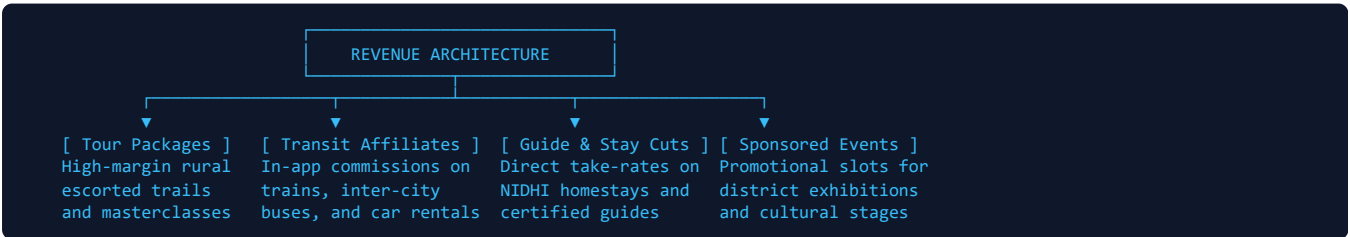
7. SYSTEM ARCHITECTURE & HIGH-LEVEL INTEGRATIONS



The SanskritiSetu system architecture operates across three distinct topological tiers:

- **External Sovereign DPI Layer:** Integrates directly with Government of India public APIs: Digital India Bhashini for offline dialect synthesis, the National Emergency Response Support System (112 NERS) for sub-cellular dispatch, NIDHI 2.0 / SAATHI for verified homestay accreditation, and ISRO Bhuvan / Survey of India for high-accuracy sovereign geospatial basemaps.
- **Backend Core Services:** Engineered on Python (FastAPI) and PostgreSQL with PostGIS extensions. Housing the Skyfield astronomical computation engine for date calculation and automated ML hygiene models for pre-triaging crowdsourced data. The dynamic bundling pipeline compiles vector tiles and embeddings into signed, versioned micro-packages.
- **Client Edge Architecture:** An on-device mobile client built with Flutter / React Native, MapLibre Native, SQLite-VSS, and ExecuTorch. It functions 100% autonomously in cellular dead zones, managing local search, offline linguistic guidance, ground-readiness calculations, and 2G binary SMS telemetry failover.

8. MULTI-STREAM BUSINESS MODEL & MONETIZATION STRATEGY



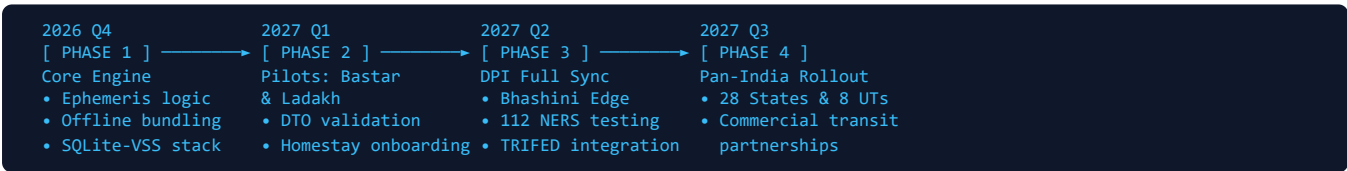
To ensure long-term commercial sustainability without compromising public-good infrastructure, SanskritiSetu implements a multi-stream monetization strategy:

- **Curated Cultural Micro-Packages:** Margin-based experiential micro-tours (e.g., 3-day Monpa weaving trails, Bastar metalcasting immersions) providing end-to-end ground transit coordination and verified village stays.
- **Transit & Travel Affiliates:** Integrated affiliate revenue on long-haul transit connecting to rural hubs (IRCTC train tickets, redBus inter-city routes, state transport corporations, and cab rentals).
- **Direct Guide & Homestay Commissions:** Standard 5–8% platform facilitation fee on pre-booked, verified rural homestays (NIDHI 2.0 / SAATHI) and certified local tour guides.
- **Artisan Commerce & Event Sponsorships:** Nominal processing commission on direct-from-artisan ODOP/TRIFED craft consignments and sponsored listing slots for regional cultural exhibitions.

9. RISK MATRIX & PRAGMATIC MITIGATIONS

RISK / FAILURE MODE	SEVERITY	LIKELIHOOD	PRAGMATIC MITIGATION STRATEGY
Dead-Zone Cash Paralysis	High	High	Cash Buffer Estimator: Pre-trip calculation of mandatory physical cash required; active alerts triggered at the last known working ATM node before entering dead zones.
Last-Mile Transit Abandonment	High	High	Transit Lifeline Nodes: Crowd-indexes informal shared sumo stands, weekly market pickup schedules, departure windows, and terrain clearance prerequisites.
Checkpoint Inadmissibility	High	Medium	Permit Navigator: Auto-generates printable, physical checkpoint logbook registration sheets adhering strictly to MHA ILP/RAP documentation standards.
DTO Portal Abandonment	Critical	High	Algorithmic Pre-Triaging: Automated anomaly checks filter spam and malformed records, reducing DTO administrative responsibilities to a fast 1-click batch review.
Over-Tourism & Ecological Strain	High	Medium	Dynamic Load-Balancing: Algorithmic prioritization that promotes secondary and tertiary festival satellites when primary circuits hit carrying-capacity thresholds.
Remote Medical Emergency	Critical	Low	Hardcoded offline directory of Primary Health Centres (PHCs) and antivenom clinics, backed by sub-cellular 112 NERS 2G binary SMS failover .

10. PHASED IMPLEMENTATION ROADMAP



- **Phase 1: Core Engine & Offline Stack (Months 1–3):**
 - Develop the astronomical ephemeris conversion engine (Skyfield).
 - Build dynamic bundling packaging architecture (<60 MB).
 - Implement SQLite-VSS embedded vector search on mobile test hardware.
 - Deliverable: Working CLI prototype and validated offline search queries.
- **Phase 2: Pilot Deployment & DTO Validation (Months 4–6):**
 - Deploy pilots across two distinct cultural testbeds: Bastar district (Chhattisgarh - tribal art/madais) and Leh district (Ladakh - monastic festivals/remote high altitude).
 - Validate the DTO 1-click batch approval workspace.
 - Ingest and sync Ministry of Tourism NIDHI 2.0 / SAATHI verified homestay data.
 - Deliverable: 100+ rural artisans and 25 homestays onboarded; DTO verification time certified under 2 minutes.
- **Phase 3: National DPI & Emergency Integration (Months 7–9):**
 - Finalize 112 NERS sub-cellular 2G binary SMS telephony integration and test failover with state dispatchers.
 - Integrate Digital India Bhashini offline speech-to-speech translation models.
 - Formalize direct digital commerce linkages with TRIFED and ODOP handicraft nodes.
 - Deliverable: Emergency dispatch integration certification and offline voice translation across 5 regional dialects.
- **Phase 4: Full Nationwide Expansion (Months 10–12):**

- Scale cultural indexing across all 28 States and 8 Union Territories.
- Onboard commercial transit affiliate APIs (IRCTC, redBus, state transport corporations).
- Official national rollout and presentation at the Ministry of Tourism annual showcase.
- Deliverable: 2,500+ traditional fairs indexed; national public accessibility.